

The factor model with uncorrelated factors is finally re-written as:

$$r = \alpha_o + Fb_f + \tilde{G}b_{\tilde{g}} + \tilde{H}b_{\tilde{h}} + \varepsilon \quad (6.34)$$

This representation now provides analysts with a methodology to calculate accurate risk exposures to a group of predefined factors which are now independent.

Estimating Covariance Using a Factor Model

A factor model across a universe of stocks can be written as:

$$R = \alpha + F\beta + \varepsilon \quad (6.35)$$

where,

$$R = \begin{bmatrix} r_{11} & r_{21} & \cdots & r_{m1} \\ r_{12} & r_{22} & \cdots & r_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1n} & r_{2n} & \cdots & r_{mn} \end{bmatrix} \quad F = \begin{bmatrix} f_{11} & f_{21} & \cdots & f_{k1} \\ f_{12} & f_{22} & \cdots & f_{k2} \\ \vdots & \vdots & \ddots & \vdots \\ f_{1n} & f_{2n} & \cdots & f_{kn} \end{bmatrix} \quad \alpha' = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$$

$$\beta = \begin{bmatrix} b_{11} & b_{21} & \cdots & b_{m1} \\ b_{12} & b_{22} & \cdots & b_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1k} & b_{2k} & \cdots & b_{mk} \end{bmatrix} \quad \varepsilon = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{21} & \cdots & \varepsilon_{m1} \\ \varepsilon_{12} & \varepsilon_{22} & \cdots & \varepsilon_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ \varepsilon_{1n} & \varepsilon_{2n} & \cdots & \varepsilon_{mn} \end{bmatrix}$$

This formulation allows us to compute the covariance across all stocks without the issues that come up when using historical market data. This process is described following Elton and Gruber (1995) as follows:

The covariance matrix of returns C is calculated as:

$$C = E[(R - E[R])'(R - E[R])]$$

From our factor model relationship we have:

$$R = \alpha + F\beta + \varepsilon$$

The expected value of returns is:

$$E[R] = \alpha + \bar{F}\beta$$

Now we can determine the excess returns as:

$$R - E[R] = (F - \bar{F})\beta + \varepsilon$$

Now substituting in the above we have:

$$\begin{aligned} C &= E[((F - \bar{F})\beta + \varepsilon)'((F - \bar{F})\beta + \varepsilon)] \\ &= E[\beta'(F - \bar{F})^2\beta + 2\beta'(F - \bar{F})\varepsilon + \varepsilon'\varepsilon] \\ &= \beta'E[(F - \bar{F})^2]\beta + 2\beta'E[(F - \bar{F})\varepsilon] + E[\varepsilon'\varepsilon] \end{aligned}$$